

```

1  # Note 10: Random Forest Algorithm
2  # =====
3
4  # Import Libraries
5  import pandas as pd
6  from sklearn.model_selection import train_test_split
7  from sklearn.ensemble import RandomForestClassifier
8  from sklearn.metrics import accuracy_score, confusion_matrix
9
10 # Load dataset
11 iris = pd.read_csv("Iris.csv")
12
13 # Features and Target
14 X = iris.drop(columns=["Species", "Id"])
15 y = iris["Species"]
16
17 # Split dataset
18 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=
42)
19
20 # Create Random Forest model
21 model = RandomForestClassifier(n_estimators=100, random_state=42)
22
23 # Train model
24 model.fit(X_train, y_train)
25
26 # Predict
27 y_pred = model.predict(X_test)
28
29 # Evaluation
30 print("\n=====")
31 print("Evaluation Report - Random Forest Classifier")
32 print("=====\n")
33 print(f"Accuracy as a Percentage : {accuracy_score(y_test, y_pred) * 100:.2f}%\n")
34 print("=====")
35 print("Confusion Matrix")
36 print("=====\n")
37 print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred), "\n")
38
39 # Predict a New Flower
40 new_flower = pd.DataFrame([[5.1, 3.5, 1.4, 0.2]], columns=["SepalLengthCm",
41 "SepalWidthCm", "PetalLengthCm", "PetalWidthCm"])
42 prediction = model.predict(new_flower)
43 print("Expected Flower : Iris-setosa")
44 print("Predicted Flower : ", prediction[0])
45
46 # Print Header
47 print("\n=====")
48 print("Important Features")
49 print("=====\n")
50
51 # Find Important Features
52 for feature, importance in zip(X_train.columns, model.feature_importances_):
    print(feature, ":", importance)

```